

Paly turns anything you want to learn into a daily retention loop.

A focused mobile MVP for students and self-learners: capture material, retrieve it actively, revisit it at the right time, and get coached through what still feels unclear.

VERSION-ONE PROMISE

From source to a trustworthy, personalized review session in under five minutes.

RECOMMENDED STACK

Mobile client + Supabase + n8n, with an AI-provider layer that can be swapped later.

MVP definition

20 September 2026

Active recall

Adaptive review

Snap-to-study

Socratic tutor

The MVP succeeds when learning becomes a closed loop.

The loop: capture a source --> extract concepts --> attempt recall --> adapt the next review --> resolve confusion with the tutor --> repeat.

Who it serves

Students need a deadline-aware path to exam readiness. **Self-learners** need a curiosity-led rhythm without the pressure of a test date. Both use the same concepts, reviews, tutor, and mastery data.

The job to be done

“Help me turn what I am learning into knowledge I can retrieve later - without spending more time organizing than learning.”

The wedge

Paly does not compete by storing more notes. It competes by converting raw material into high-quality retrieval practice and knowing what the learner should do next.

Ship in version one

- Exam Mode and Explore Mode onboarding
- Text, photo, and PDF capture
- AI-generated concepts and recall prompts
- Editable study sets with source anchors
- Adaptive daily review queue
- Grounded text and push-to-talk tutor
- Readiness / mastery dashboard
- Due-review notifications

Do not ship yet

Public profiles, social feeds, study-room avatars, creator marketplaces, competitive leaderboards, institution dashboards, complex note editing, or a desktop authoring suite.

Decision rule

Every v1 feature must improve one of three moments: turning material into practice, completing effective retrieval, or returning when a review is due. If it does not, defer it.

Two modes share one learning engine.

Mode changes the plan and the emotional framing. It does not fork the underlying data, study items, scheduler, or tutor.

Deadline-driven

Exam Mode

Input: exam date, subject, available study days, optional syllabus or weighting.

Home: today's plan, due reviews, uncovered topics, readiness trend.

Planning: work backwards from the deadline; protect time for mixed review and practice exams.

Feedback: "You are strong on cell structure; energy transfer needs two more successful recalls before Friday."

Curiosity-driven

Explore Mode

Input: topic, motivation, preferred cadence, optional outcome.

Home: next useful session, active topics, growing knowledge map, momentum.

Planning: balance due reviews with one small piece of new material.

Feedback: "Your mental model of neural networks is holding. Next, connect activation functions to gradient flow."

Shared product principles

Recall before reveal. Ask for an answer before showing it. Reading is support; retrieval is the core interaction.

One next action. The home screen should answer "What should I study now?" without requiring planning.

Confidence is not mastery. Capture how sure the learner felt, but update mastery primarily from performance and time.

Show the source. Every AI-generated concept, question, and tutor answer can point back to the learner's material.

Correct before saving. The learner can edit generated content; uncertainty is flagged instead of hidden.

Frictionless sessions. A useful review can be completed in two minutes, while longer sessions remain possible.

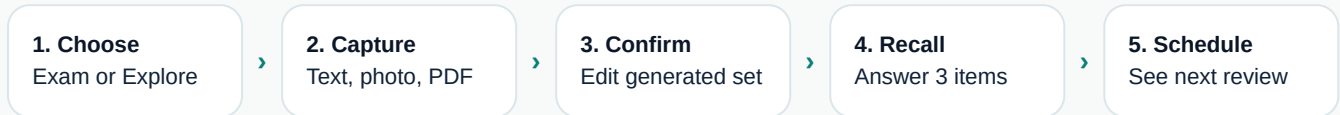
Seven capabilities complete the core loop.

CAPABILITY	VERSION-ONE BEHAVIOR	DONE WHEN
1. Goal setup	Choose Exam or Explore; define subject, outcome, cadence, and optional deadline.	A first session is planned.
2. Capture	Paste text, photograph pages or a whiteboard, or import a PDF. Show extraction quality.	Readable source text is stored privately.
3. Study-set generation	Extract concepts, concise explanations, and recall prompts. Require preview and edit before publish.	Each item links to a source passage.
4. Active recall	Short answer, cloze, and explain-back prompts. Reveal answer, capture result and confidence.	Every attempt creates a review record.
5. Adaptive review	Order due items by urgency, weakness, goal relevance, and available time; calculate the next due date.	The queue adapts after each answer.
6. Socratic tutor	Text plus push-to-talk. Ask guiding questions, detect gaps, and cite the learner's sources.	A tutor turn can become a study item.
7. Progress + return	Mastery by concept, exam readiness or learning momentum, session recap, due-review alerts.	The learner knows what changed and what is next.

Scope guard: “AI generated it” is not acceptance. The output must be editable, traceable to source material, and converted into a measurable learning action.

The shortest successful journey ends with a scheduled review.

Activation journey



Capture acceptance criteria

- Camera supports crop, rotate, retake, and multi-page capture.
- Optical character recognition output is shown when confidence is low.
- PDF import preserves page references.
- Generation can be retried without creating duplicate sources or items.
- The learner can remove irrelevant passages before generation.

Study acceptance criteria

- Answer is hidden until an attempt or “I don’t know.”
- Rating choices: Again, Hard, Good, Easy.
- Keyboard, tap, and voice answers all resolve to one review model.
- Session resumes after interruption.
- Recap names improved and weak concepts.

Tutor acceptance criteria

- Starts by checking what the learner already believes.
- Asks one question at a time.
- Uses hints before giving the full answer.
- References source passages used in its response.
- Offers “turn this into a card” after resolving a gap.
- Voice has a visible transcript and a text fallback.

Return acceptance criteria

- Notifications state the useful action, not generic streak pressure.
- Quiet hours and frequency are user-controlled.
- Tapping an alert opens the due queue directly.
- Missed days reflow the plan without punishment.

Retention is a state machine, not a streak.

Every published study item carries a memory state. A review changes that state and produces the next due date. The exact scheduling formula can evolve without changing the product contract.

D**Difficulty**

How hard this item tends to be for this learner.

S**Stability**

How long the memory is likely to remain retrievable.

R**Retrievability**

Estimated chance of successful recall now.

Review update

```
input: item_state, result, confidence, response_time, reviewed_at

if result == AGAIN:
    reduce stability; increase urgency; schedule soon
else:
    adjust difficulty; extend stability by result strength

next_due = reviewed_at + interval(item_state, goal_pressure)
write review event; update item state atomically
```

Queue score

Priority = overdue urgency + weakness + goal relevance + novelty balance - recent exposure.

Exam Mode can increase goal relevance as the deadline approaches. Explore Mode can reserve a small share of the session for new concepts.

Mastery display

Show concept states such as New, Learning, Holding, and Strong. Avoid false precision. An 84% score implies certainty the model does not have.

Readiness combines coverage, due load, and successful mixed recall; it is a planning signal, not a grade prediction.

The tutor teaches from the learner's sources.

Question
Text or voice



Retrieve
Relevant source chunks



Coach
Probe, hint, explain



Learn
Record gap or create item

Tutor policy

1. Identify the concept and the learner's stated goal.
2. Retrieve a small, relevant source context.
3. Ask for the learner's current explanation.
4. Probe one gap at a time; prefer hints to answers.
5. Correct clearly when a misconception persists.
6. End with a concise explain-back check.

Guardrails

- Label when an answer goes beyond uploaded material.
- Never pretend a source supports a claim it does not.
- Preserve page or passage anchors in citations.
- Do not automatically publish generated cards.
- Store transcript only with clear user controls.
- Cap voice sessions by plan to control latency and cost.

Generation contract

OUTPUT	REQUIRED FIELDS	VALIDATION
Concept	Name, explanation, source anchors	Unique within goal; evidence exists
Recall item	Prompt, expected answer, type, concept IDs	Answerable from source; no duplicates
Tutor reply	Response, next question, cited chunks	Citations resolve; one question max
Session recap	Wins, gaps, next actions	Derived from review events only

Separate content, memory, and activity.

The schema uses **auth.users.id** as the ownership root. All user-owned tables carry **user_id** for simple row-level security.

TABLE	KEY FIELDS	PURPOSE
profiles	user_id, display_name, timezone, locale, preferences_json	Learner settings and notification defaults
learning_goals	id, user_id, mode, title, exam_at, cadence, status	Exam or Explore container
sources	id, goal_id, type, title, storage_path, status, page_count	Uploaded or pasted learning material
source_chunks	id, source_id, ordinal, text, page_ref, embedding, checksum	Retrieval units with stable provenance
concepts	id, goal_id, name, explanation, state, archived_at	Canonical knowledge units
concept_sources	concept_id, chunk_id, relevance	Evidence links for explanations
concept_edges	from_concept_id, to_concept_id, relation	Prerequisite and related-concept graph

Lineage rule: a published concept must have at least one source-chunk link. If the learner creates it manually, mark its origin as user-authored rather than inventing a citation.

Every answer becomes durable learning history.

TABLE	KEY FIELDS	PURPOSE
study_items	id, goal_id, prompt, answer, item_type, origin, status	Editable recall prompts
item_concepts	item_id, concept_id, weight	Many-to-many skill mapping
memory_states	item_id, user_id, difficulty, stability, due_at, last_review_at, lapses	Current scheduler state
study_sessions	id, user_id, goal_id, mode, started_at, ended_at, planned_minutes	One learning session
reviews	id, session_id, item_id, result, confidence, response_ms, answer_text, reviewed_at	Immutable recall event
study_plans	id, goal_id, version, window_start, window_end, status	Versioned plan snapshot
plan_items	plan_id, concept_id, due_on, activity_type, minutes, completed_at	Daily planned work
tutor_threads	id, goal_id, concept_id, started_at, status	Grounded coaching context
tutor_messages	id, thread_id, role, content, audio_path, created_at	Transcript and voice references
message_citations	message_id, chunk_id, quote_start, quote_end	Traceable tutor evidence

Essential relationships

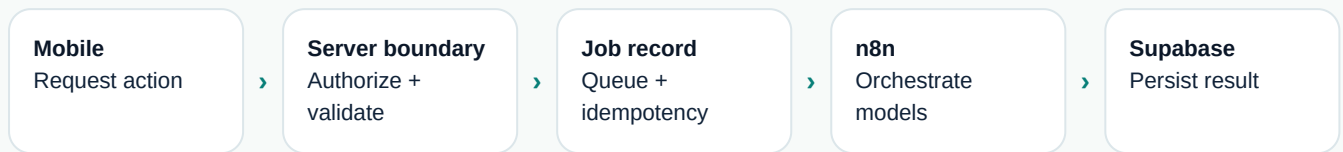
```

auth.users 1--1 profiles
profiles 1--* learning_goals 1--* sources 1--* source_chunks
learning_goals 1--* concepts *--* study_items
study_items 1--1 memory_states 1--* reviews
study_sessions 1--* reviews
concepts 1--* tutor_threads 1--* tutor_messages
source_chunks *--* concepts; source_chunks *--* tutor_messages

```

Use UUID primary keys, UTC timestamps, soft archive for learner content, and immutable review events. Store files in private Supabase Storage; keep only paths in tables.

n8n jobs should be asynchronous and observable.



WORKFLOW	INPUT	OUTPUT
INGEST_SOURCE	source_id, file path	Text, chunks, quality state
GENERATE_SET	goal_id, chunk IDs, preferences	Draft concepts and items
TUTOR_TURN	thread_id, message, mode	Reply, citations, next question
SESSION_RECAP	session_id	Wins, gaps, next actions
REBUILD_PLAN	goal_id, reason	New plan version

ai_jobs

id, user_id, job_type, entity_id, idempotency_key, status, attempt_count, provider, error_code, requested_at, completed_at.

Webhook rule

The app never holds an n8n secret or Supabase service-role key. A trusted server boundary signs outbound requests and validates callbacks.

Privacy and observability belong in the MVP.

Required security controls

- Row-level security on every user-owned table.
- Private storage buckets with short-lived signed access.
- Service-role credentials only in trusted server environments.
- Ownership checks before any job runs or callback writes.
- Delete-account flow that removes files, content, and transcripts.
- Rate limits and file-size limits at the server boundary.
- Do not train models on learner content unless they explicitly opt in.

Cost controls

- Chunk and hash sources so unchanged content is not processed twice.
- Use smaller models for extraction and classification where quality holds.
- Limit context to the most relevant chunks.
- Set voice duration and generation quotas by plan.
- Track cost per activated learner and per completed session.

Events to instrument

```
goal_created
source_uploaded
source_ingested
study_set_generated
study_set_published
session_started
review_answered
session_completed
tutor_turn_completed
study_item_created_from_tutor
notification_opened
plan_rebuilt
```

Beta scorecard

Activation: a learner publishes a set, answers at least three items, and receives a next due date.

Learning: improvement in successful delayed recall for repeated items.

Habit: due reviews completed and active weeks, not raw app opens.

Trust: edits, source-open rate, tutor citation use, and “not helpful” reports.

Efficiency: time from upload to first review and AI cost per completed session.

Recommended beta threshold: set numeric targets only after observing the first 25-50 learners. Until then, optimize for completed retention loops and investigate every failed activation.

Build the loop before the ecosystem.

Milestone 0 · Foundation

Auth, profiles, goals, database migrations, private storage, job table, analytics events, design system.

Milestone 3 · Tutor

Grounded retrieval, Socratic policy, citations, text chat, push-to-talk, card creation from a gap.

Milestone 1 · Capture

Text/photo/PDF intake, ingestion workflow, source viewer, chunks, quality states, retry and deduplication.

Milestone 4 · Planning

Exam readiness, Explore momentum, plan rebuild, notifications, quiet hours, missed-day recovery.

Milestone 2 · Recall

Generated draft sets, editor, study session, review events, memory state, daily queue, recap.

Milestone 5 · Closed beta

Cost dashboard, failure triage, deletion checks, accessibility pass, onboarding interviews, release gates.

Launch gates

AREA	MINIMUM BAR
Core loop	A new learner can reach a scheduled review without assistance.
Reliability	Jobs are retry-safe; failures are visible and recoverable.
Trust	Generated claims link to source passages; edits persist.
Learning	Review history changes the next queue and due dates.
Privacy	Cross-user access tests fail; account deletion is complete.
Accessibility	Core flows support screen readers, dynamic text, and non-color cues.

First build decision: prototype the full path with one source type and text tutoring first. Once the data contracts hold, add PDF/photo intake and voice without changing the learning model.